

Engineering Curves: Construction of Hypocycloid

Rolling Circle Method with Tangent & Normal

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Problem Statement

Problem

Construct one complete hypocycloid arch generated by a circle of diameter 30 mm rolling internally, without slipping, inside a fixed/directing circle of diameter 150 mm. The starting point of contact is A , the angle $AOB = 120^\circ$, and the arch is to be divided into 12 equal angular parts. Construct the successive positions of the generating circle and tracing point, and construct the tangent and normal at a selected point P on the hypocycloid.

Given Data and Calculations

Given data

Diameter of directing circle = 150 mm,

$$R = 75 \text{ mm},$$

Diameter of generating circle = 30 mm,

$$r = 15 \text{ mm},$$

$$\angle AOB = 120^\circ,$$

Let number of divisions = 12.

Calculations

$$OC_0 = R - r = 75 - 15 = 60 \text{ mm},$$

$$AC_0 = r = 15 \text{ mm},$$

$$\Delta\theta = \frac{120^\circ}{12} = 10^\circ.$$

Generating-circle division

$$\text{Division angle} = \frac{360^\circ}{12} = 30^\circ.$$

Rolling relation

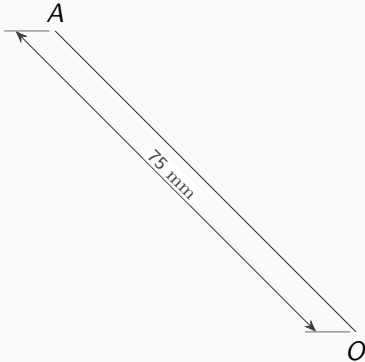
Each 10° centre step $\leftrightarrow$ 30° generating-circle step.

Centre locus

$$OC = R - r = 60 \text{ mm}.$$

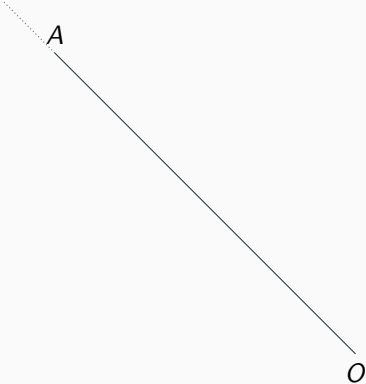
Step 1: Draw OA

$OA = 75 \text{ mm.}$



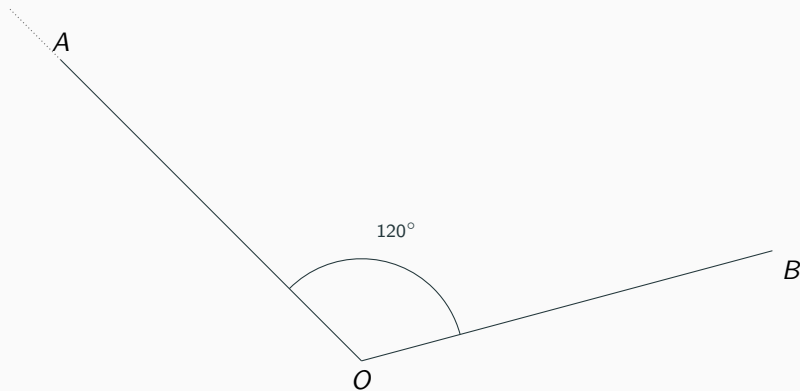
Step 2: Extend OA

Extend OA .



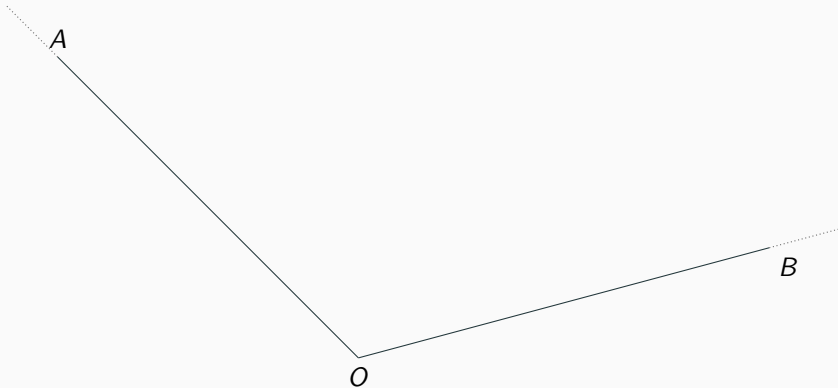
Step 3: Draw OB at 120°

$$\angle AOB = 120^\circ.$$

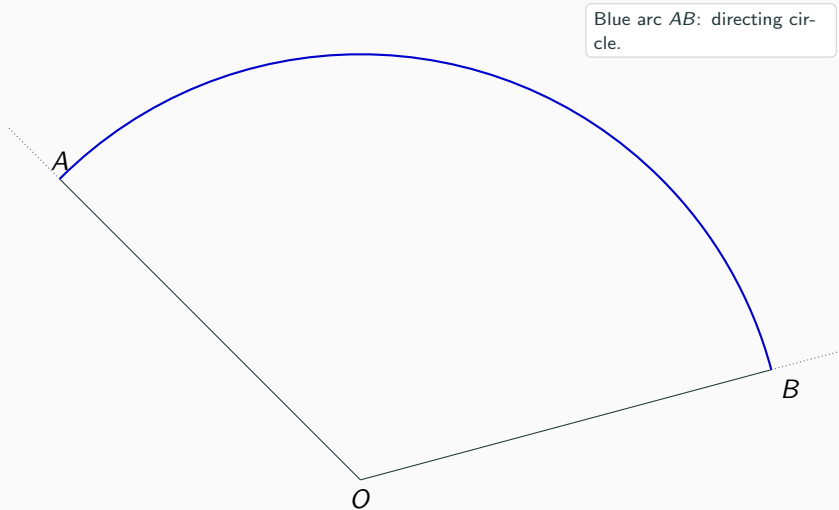


Step 4: Extend OB

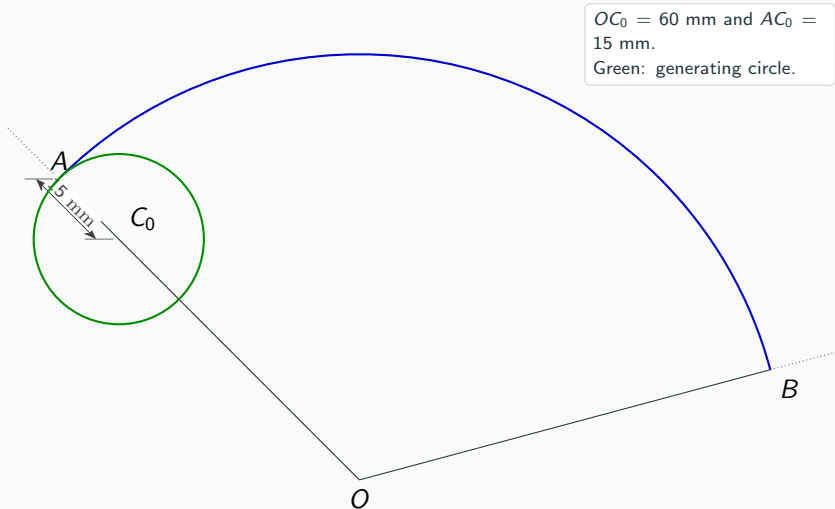
Extend OB .



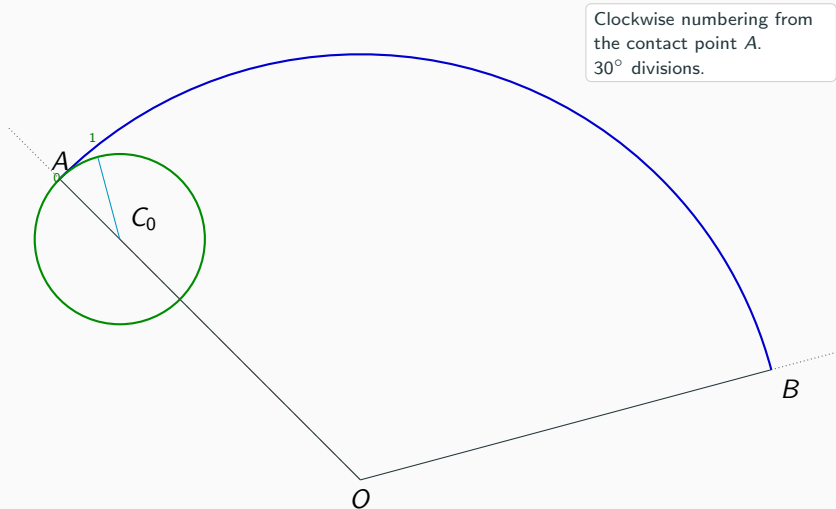
Step 5: Draw the directing arc AB



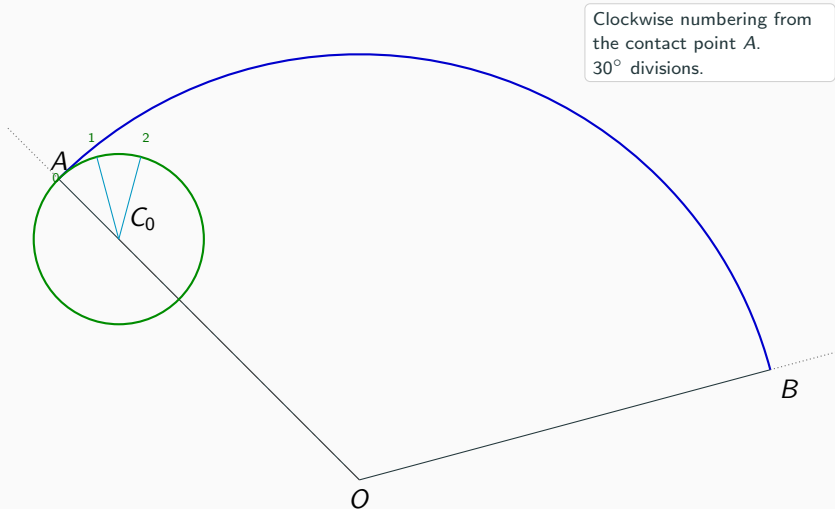
Step 6: Locate C_0 and the generating circle



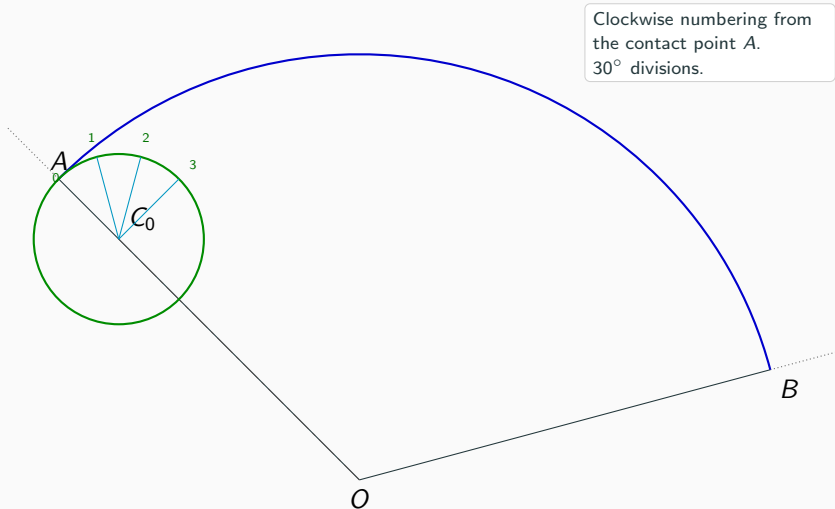
Step 7: Divide the generating circle into 12 equal parts



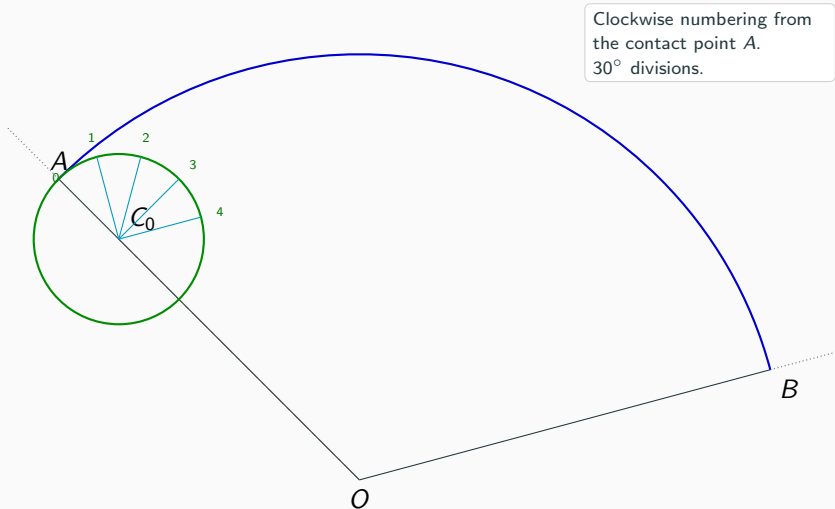
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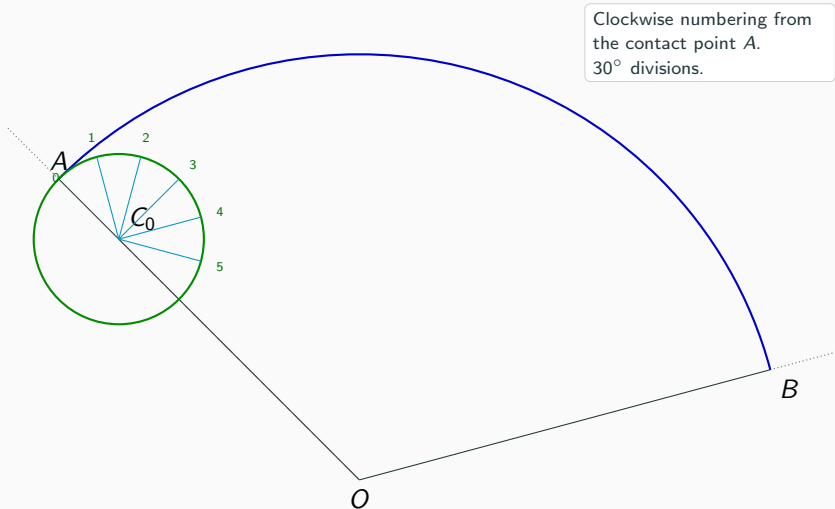
Step 7: Divide the generating circle into 12 equal parts



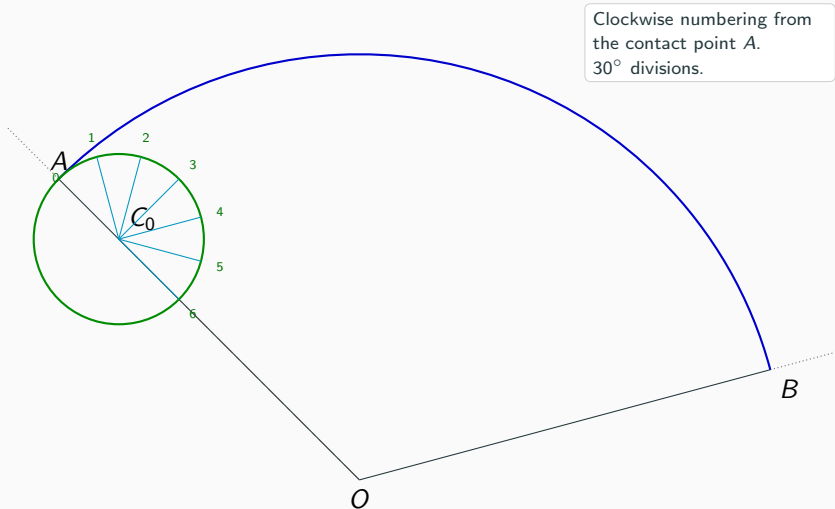
Step 7: Divide the generating circle into 12 equal parts



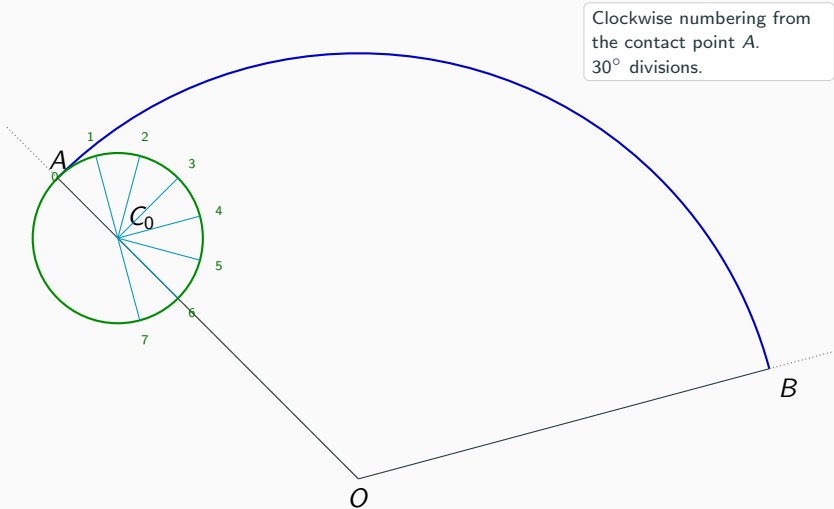
Step 7: Divide the generating circle into 12 equal parts



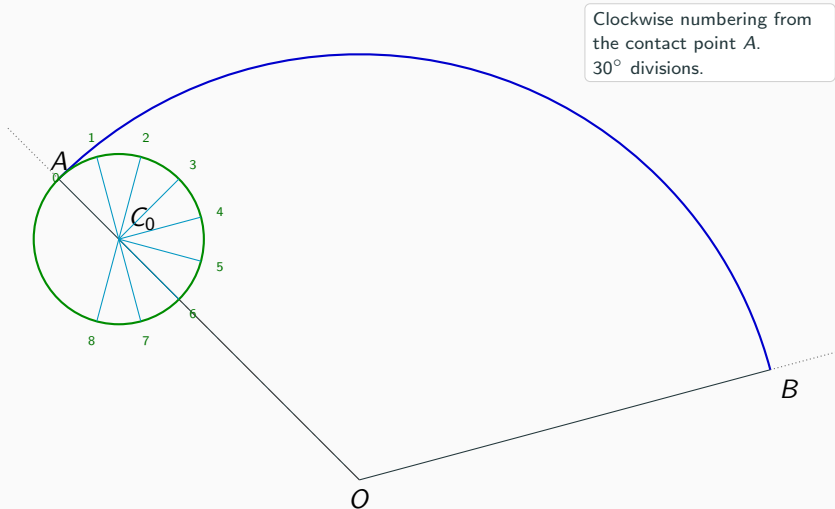
Step 7: Divide the generating circle into 12 equal parts



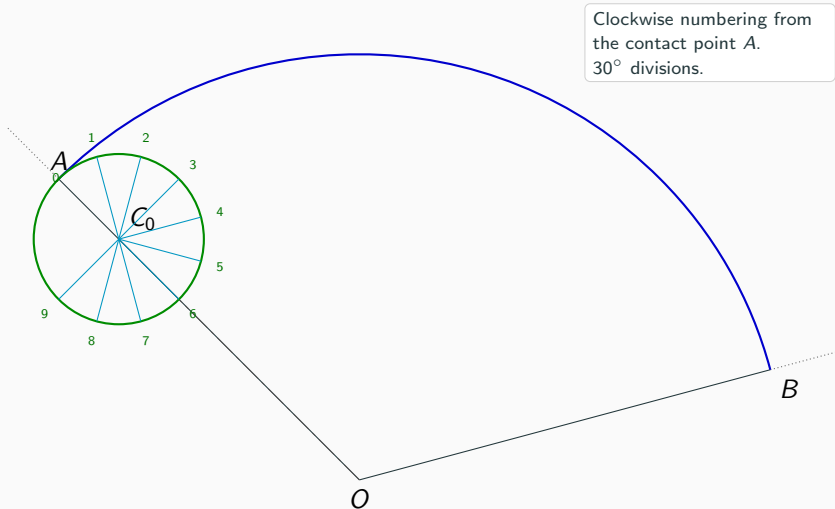
Step 7: Divide the generating circle into 12 equal parts



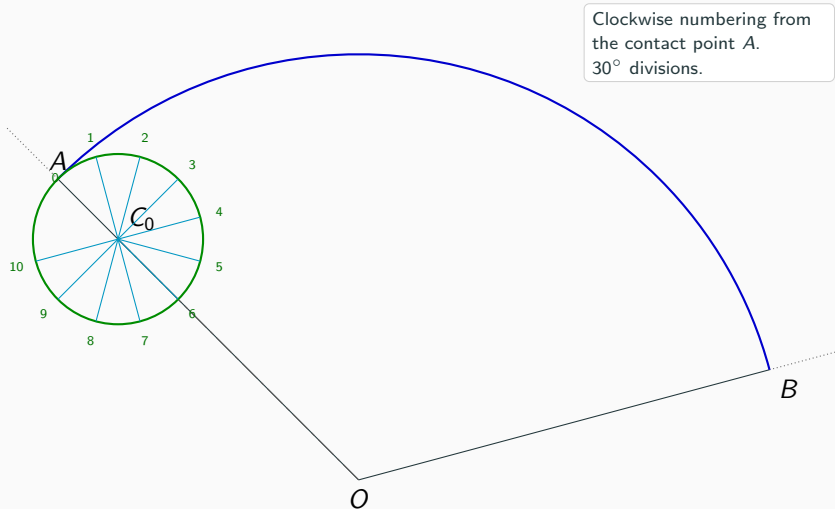
Step 7: Divide the generating circle into 12 equal parts



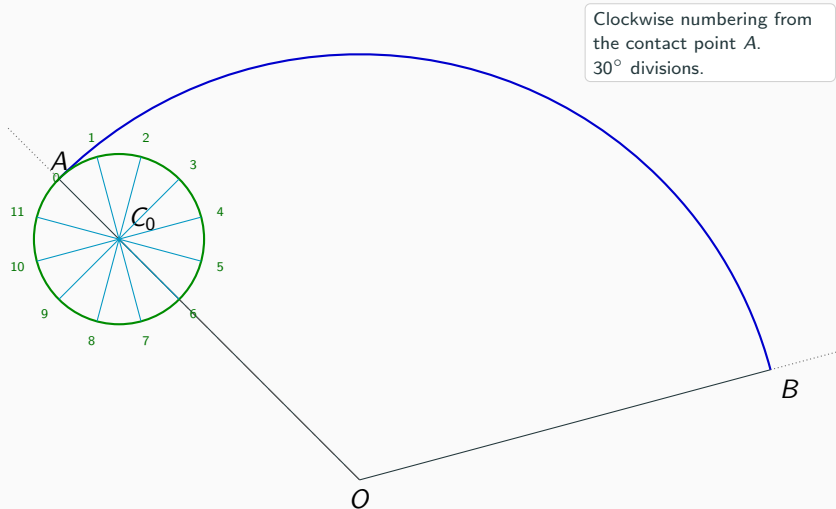
Step 7: Divide the generating circle into 12 equal parts



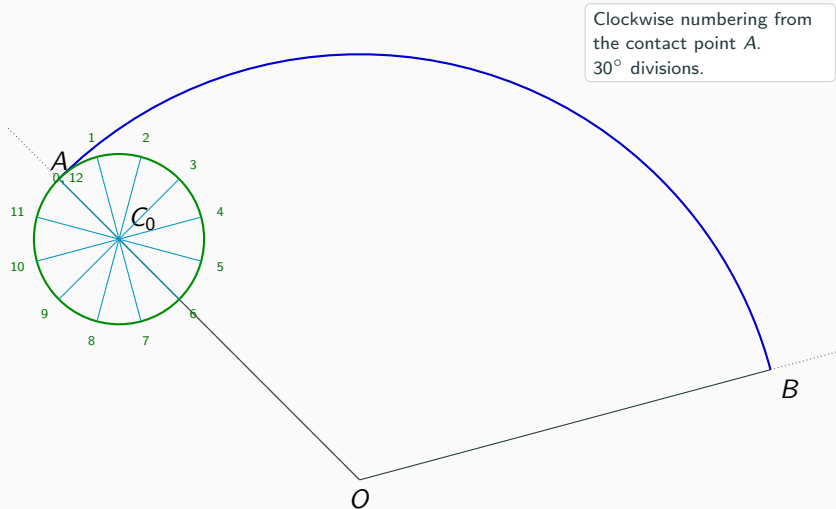
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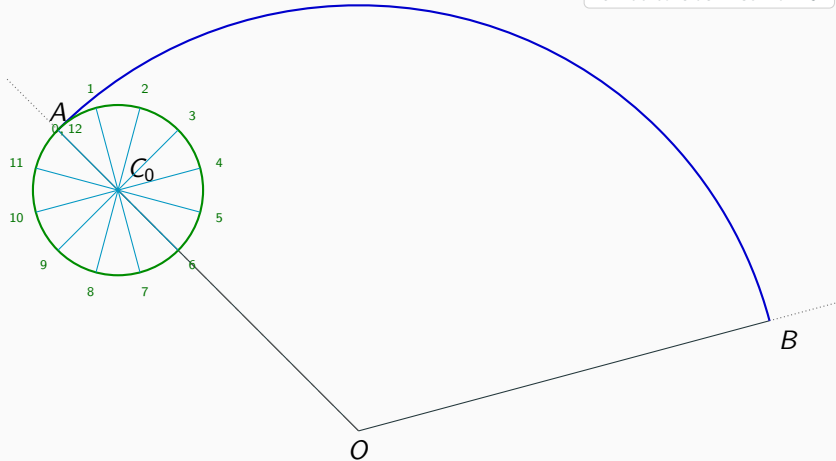


Step 7: Divide the generating circle into 12 equal parts

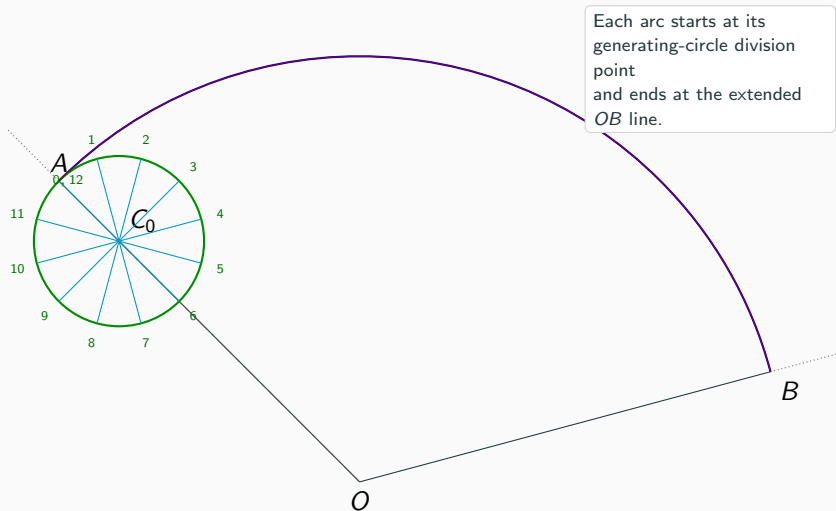


Step 8: Complete the generating-circle divisions

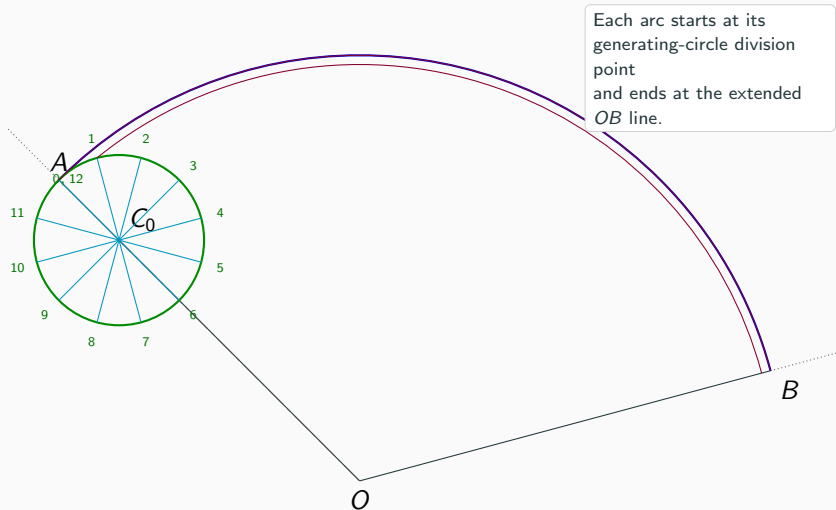
Generating-circle divisions
numbered clockwise from 0.



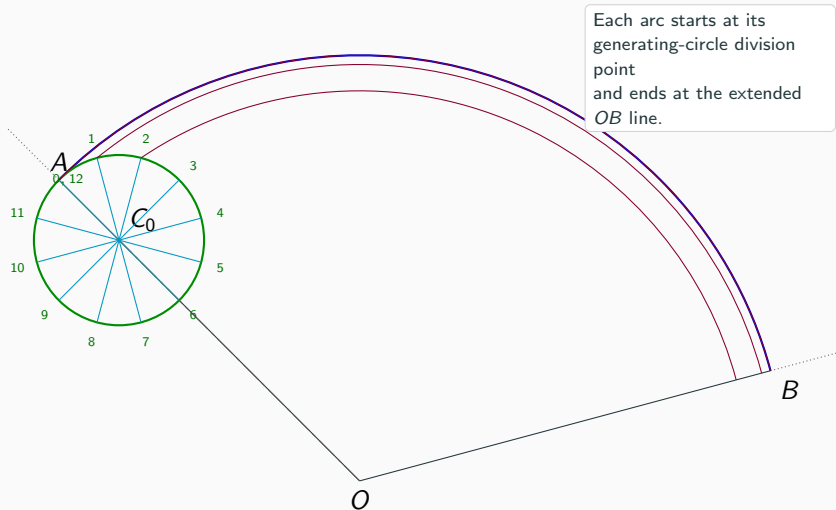
Step 9: Draw concentric construction arcs



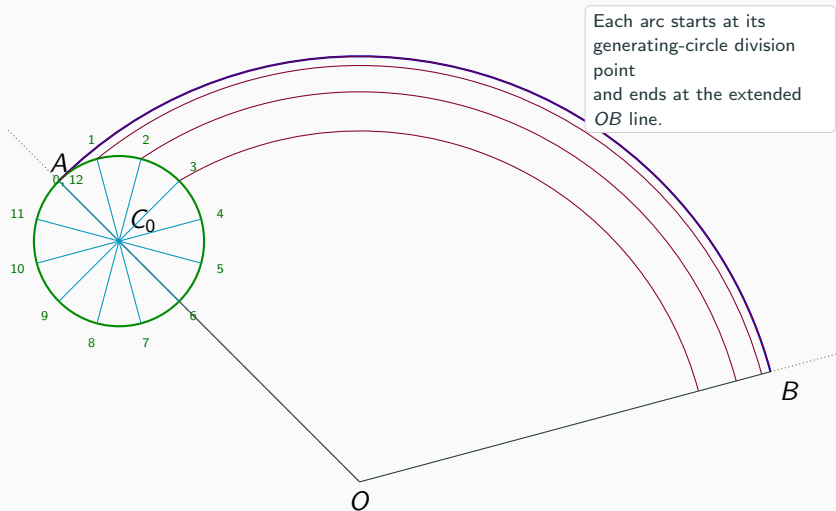
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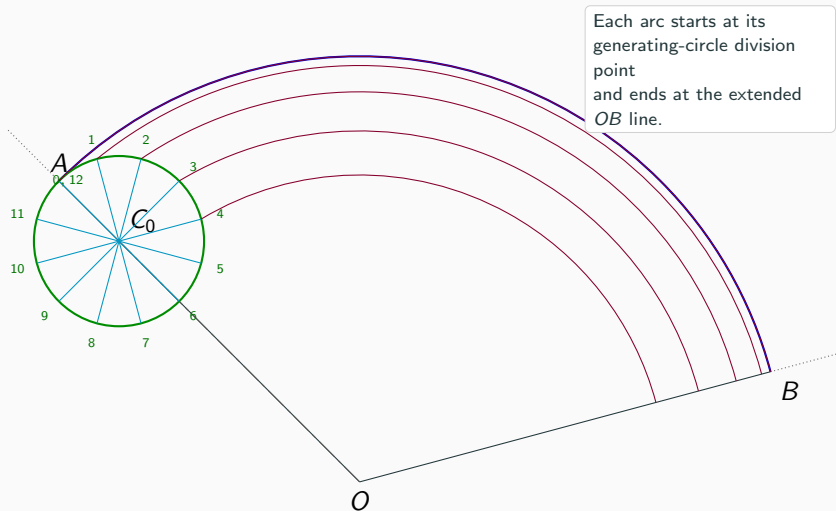
Step 9: Draw concentric construction arcs



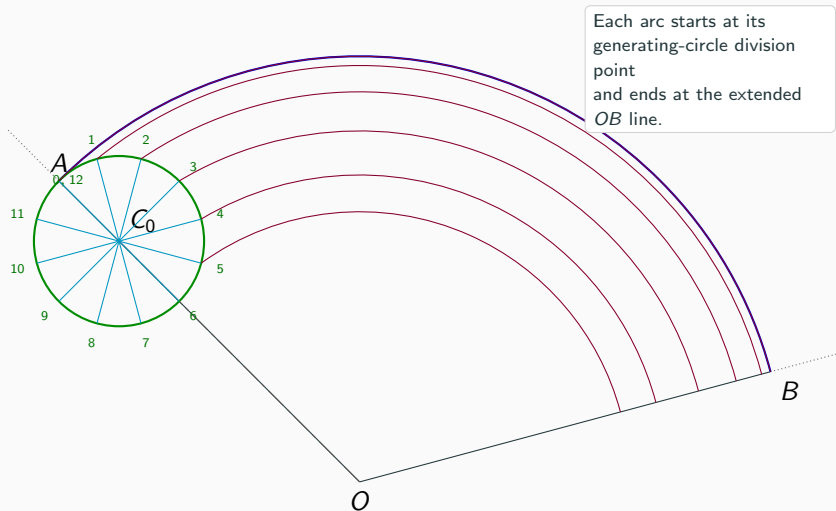
Step 9: Draw concentric construction arcs



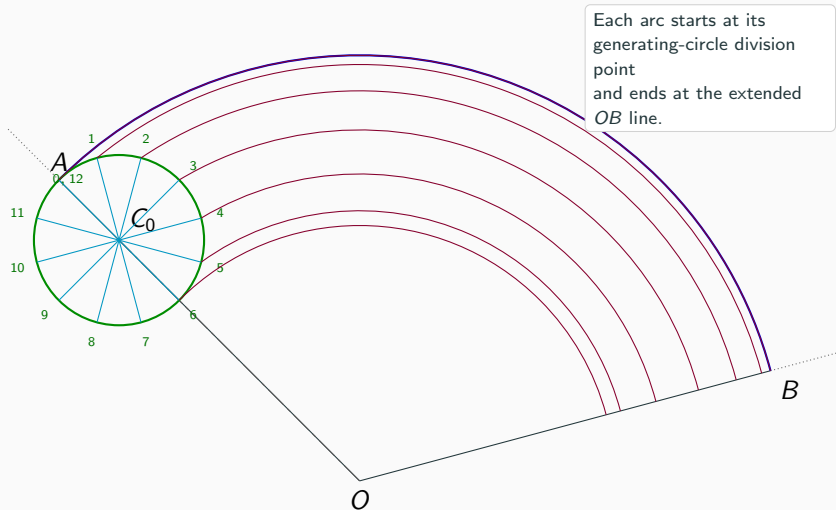
Step 9: Draw concentric construction arcs



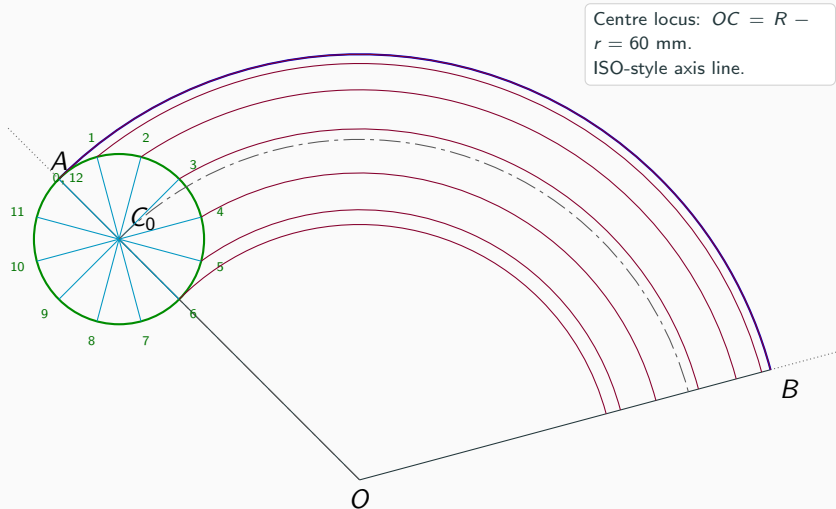
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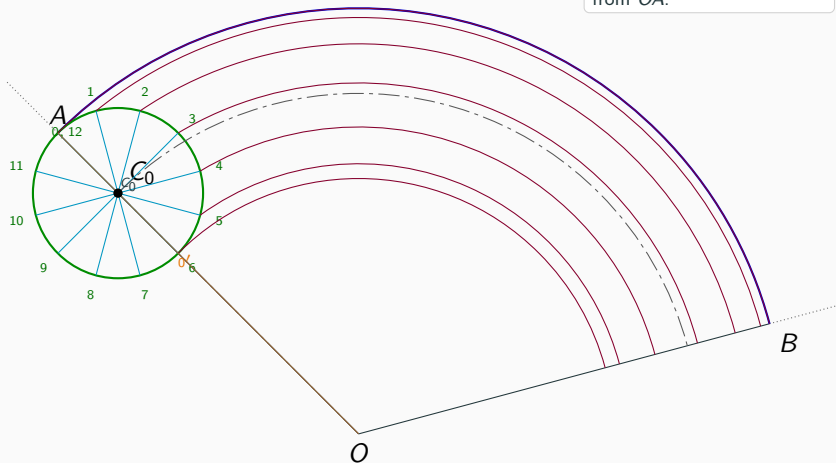


Step 10: Draw the centre arc through C_0



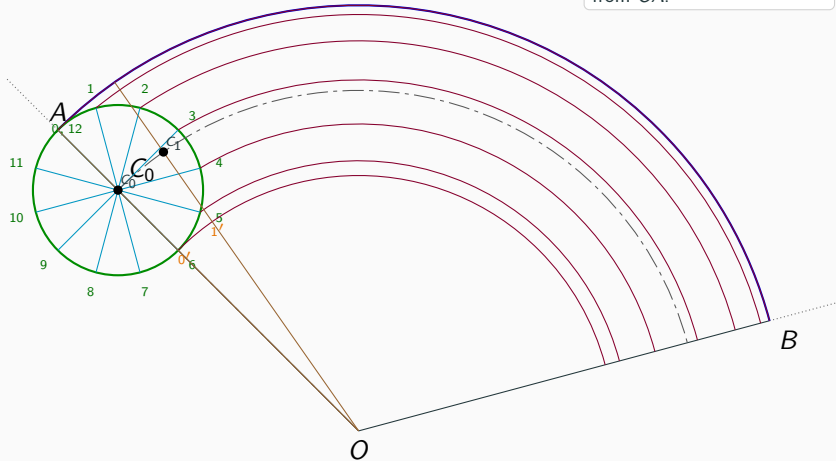
Step 11: Divide $\angle AOB$ into 12 equal parts

10° divisions, clockwise
from OA.



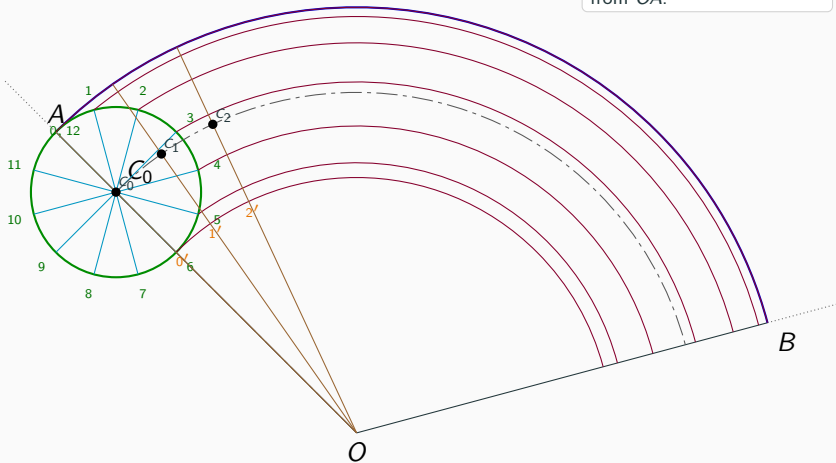
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10° divisions, clockwise
from OA .



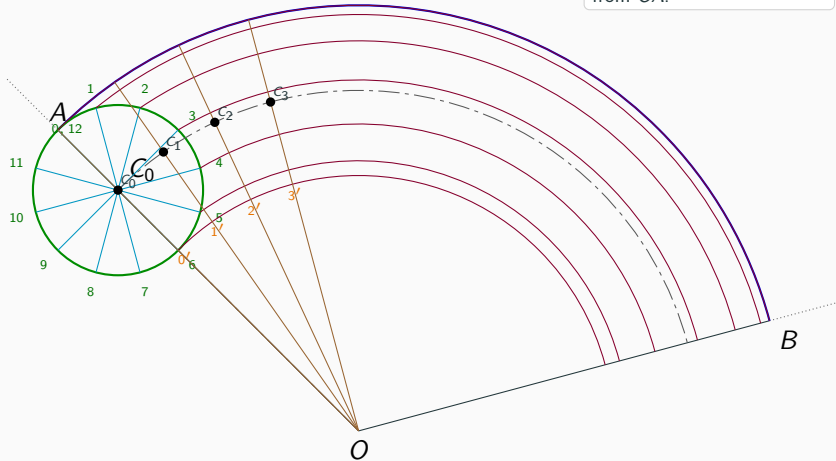
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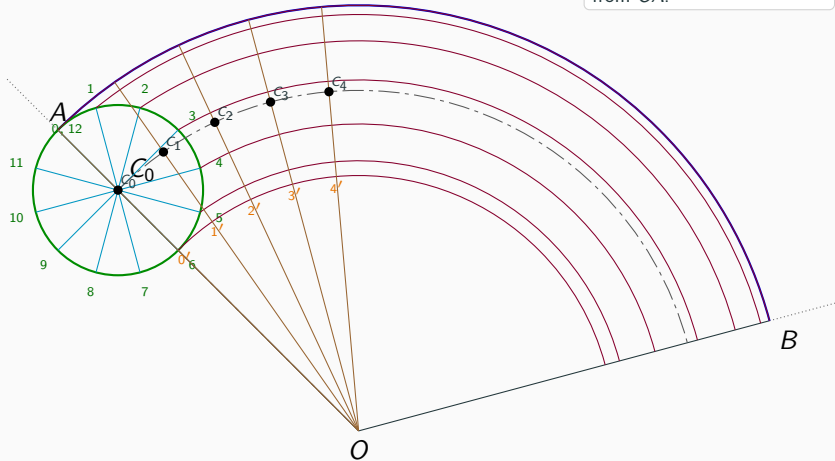
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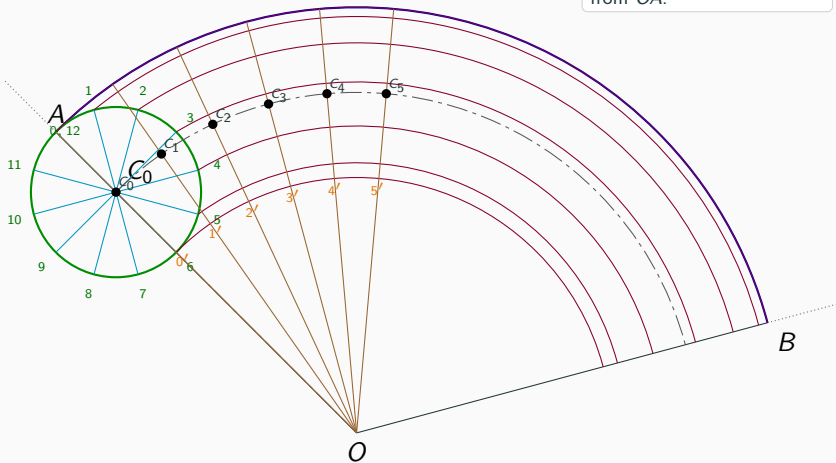
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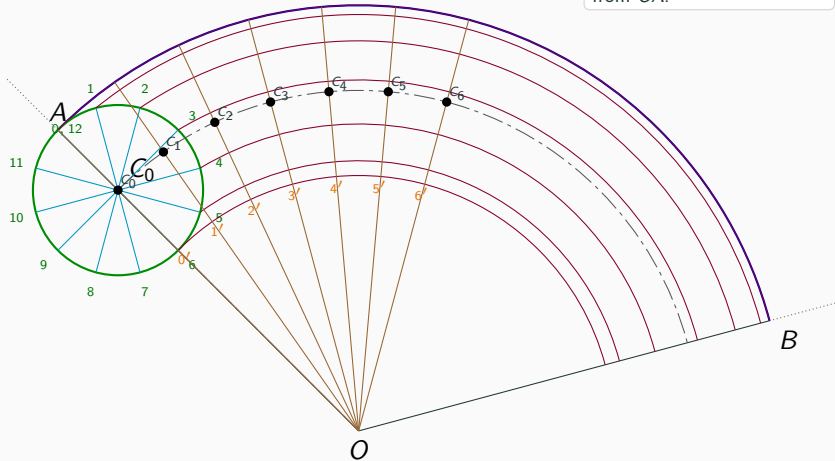
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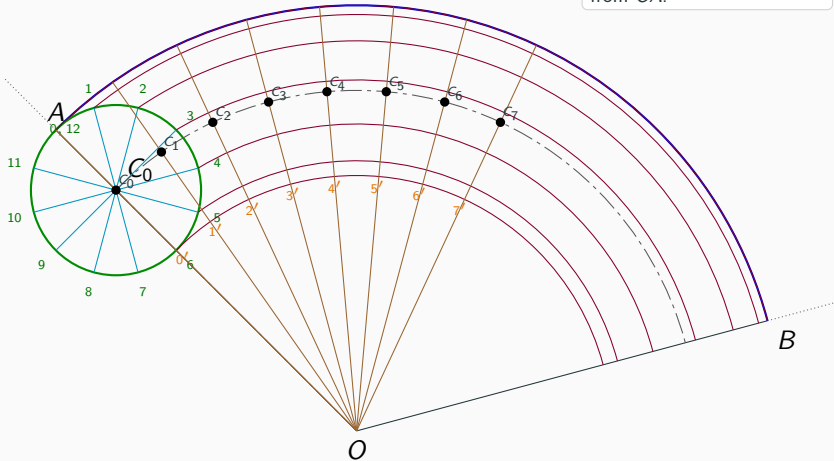
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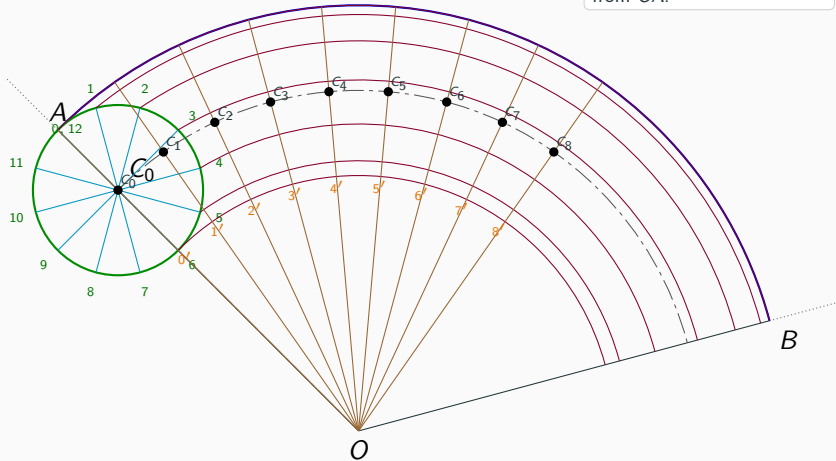
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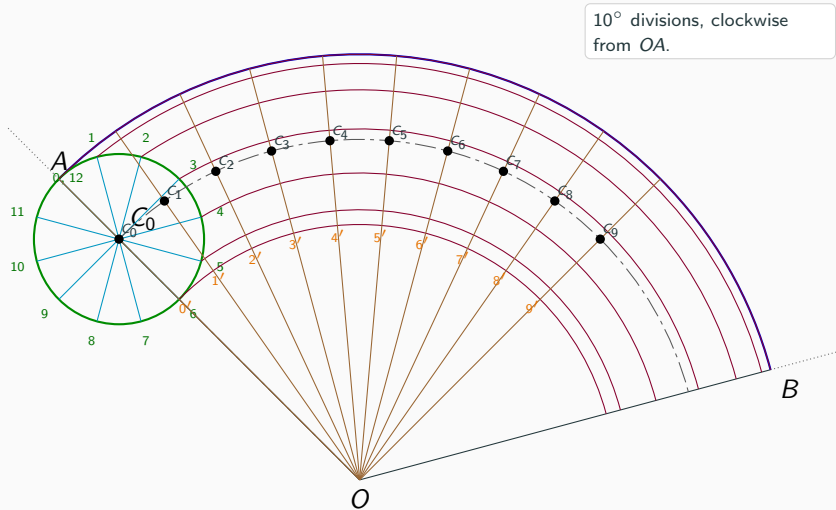


Step 11: Divide $\angle AOB$ into 12 equal parts

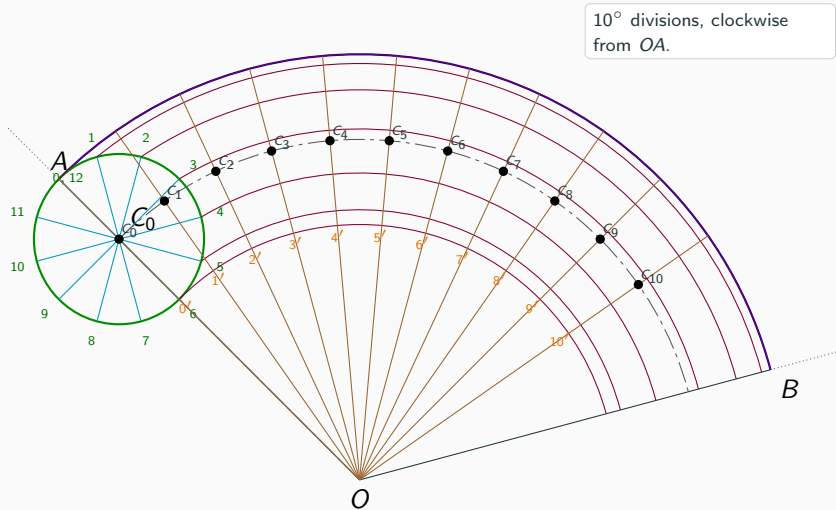
10° divisions, clockwise
from OA .



Step 11: Divide $\angle AOB$ into 12 equal parts

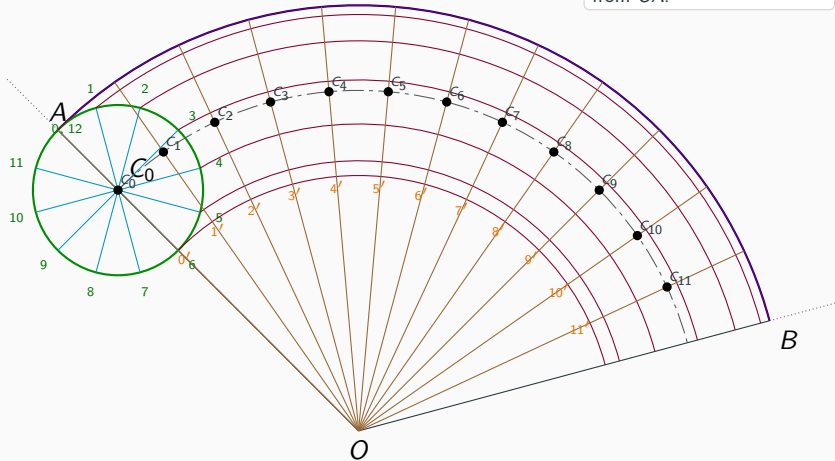


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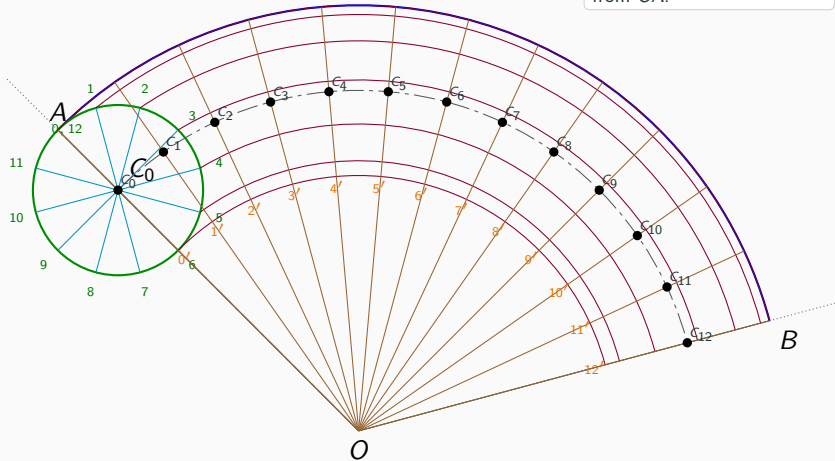
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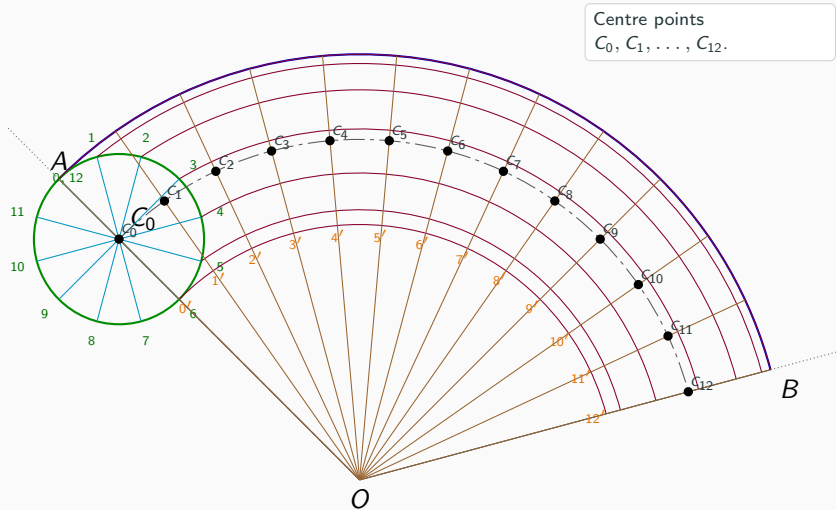


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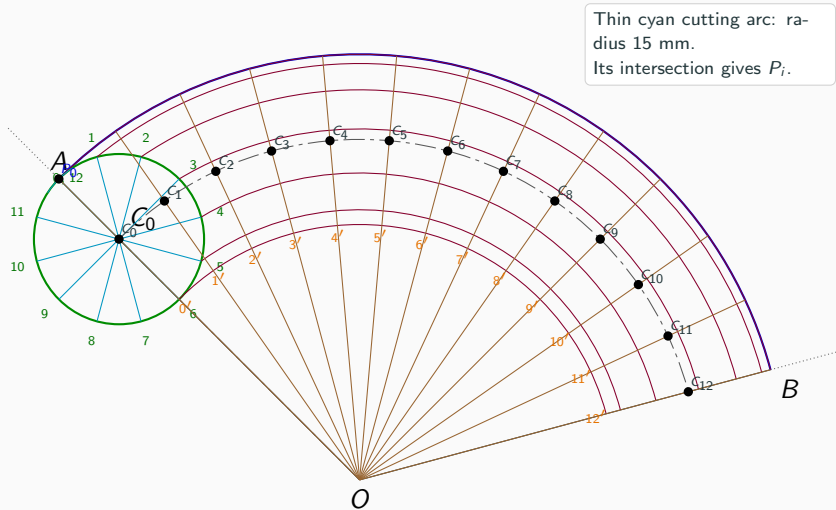
10° divisions, clockwise
from OA .



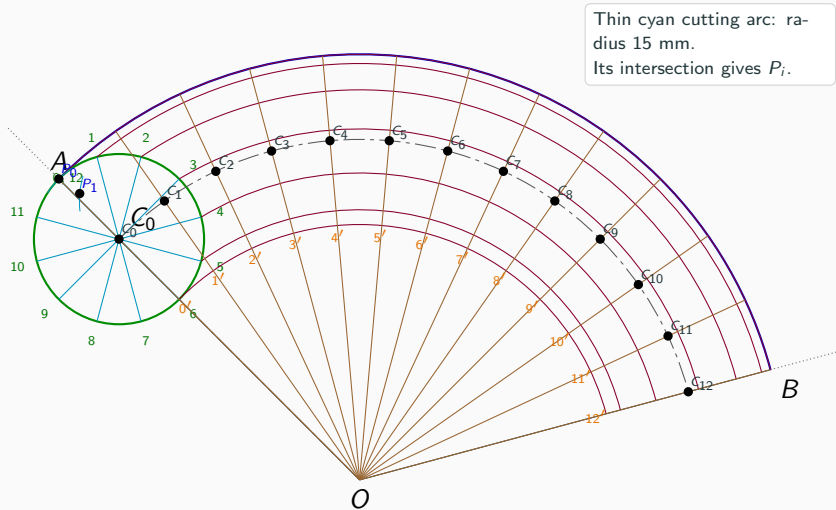
Step 12: Complete the angular divisions and centre points



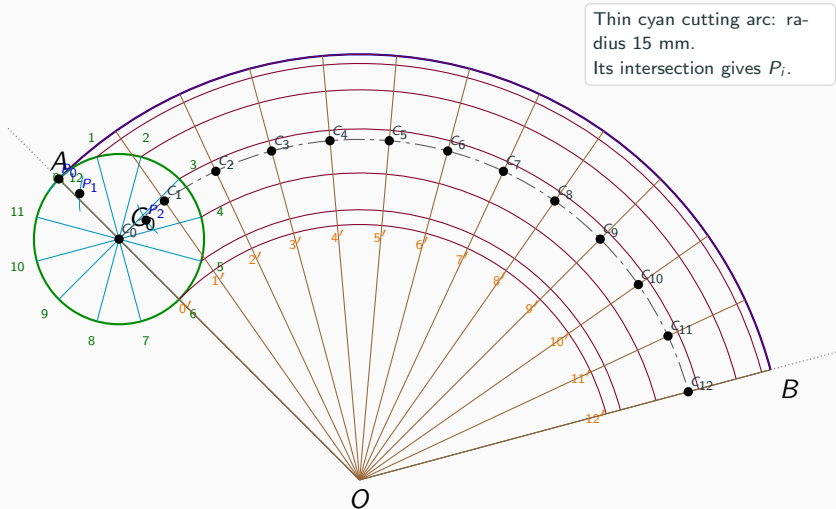
Step 13: Draw the small cutting arc and locate P_i



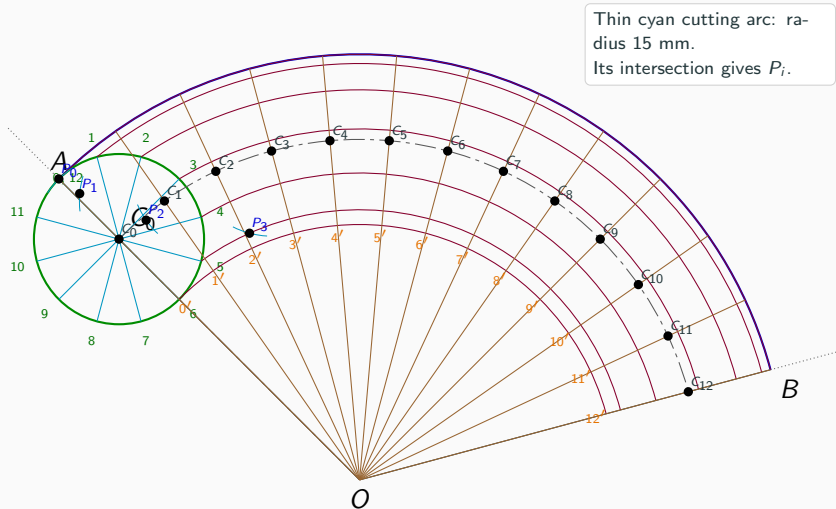
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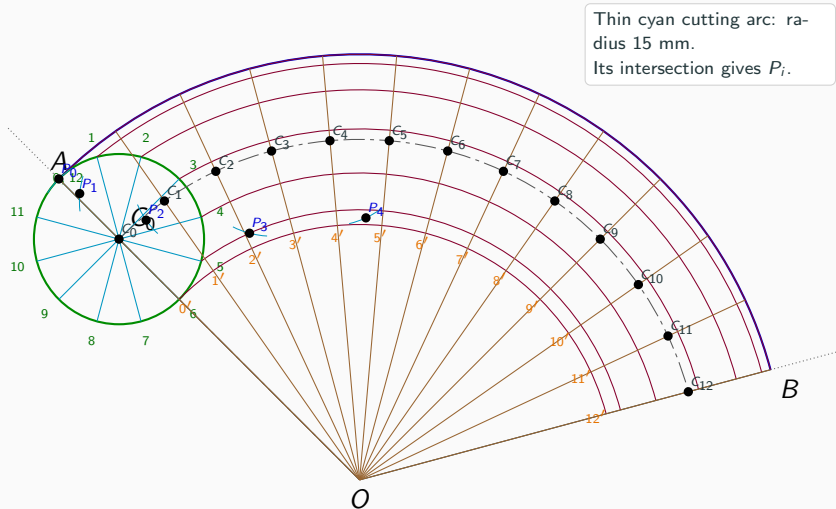
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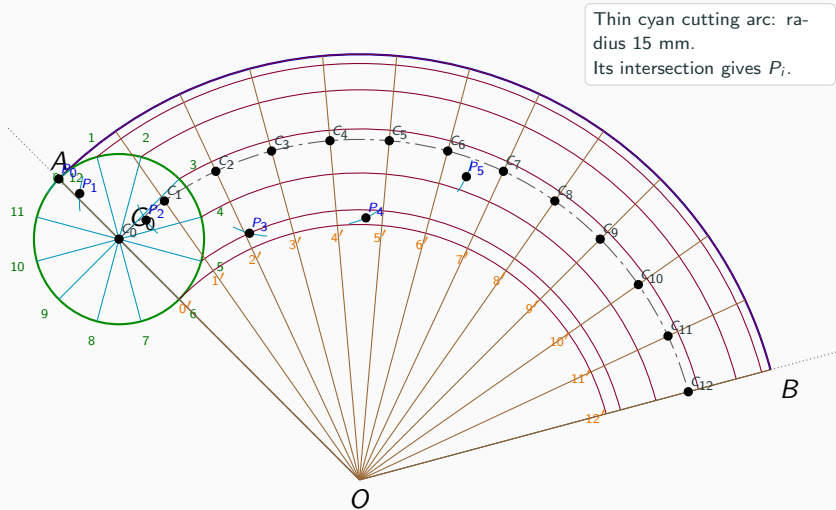
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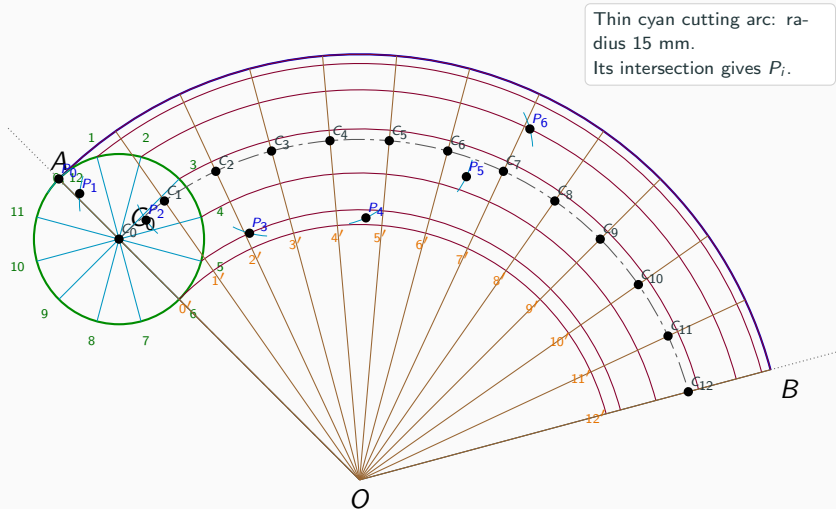
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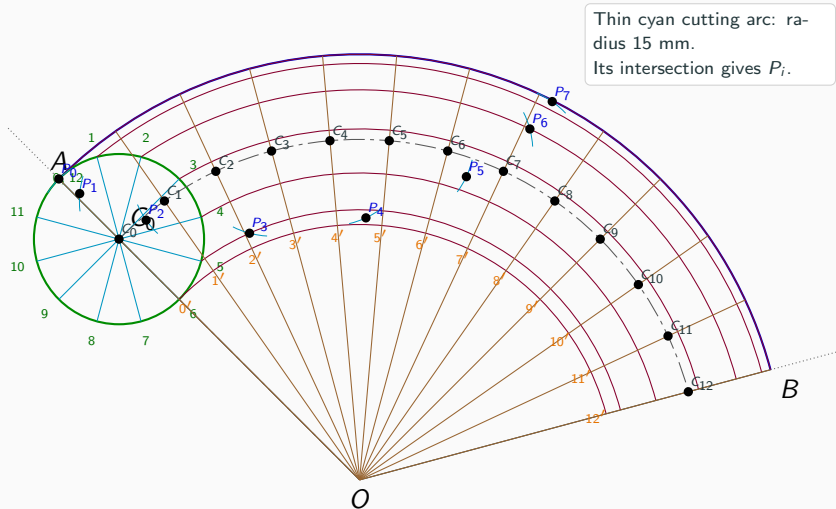
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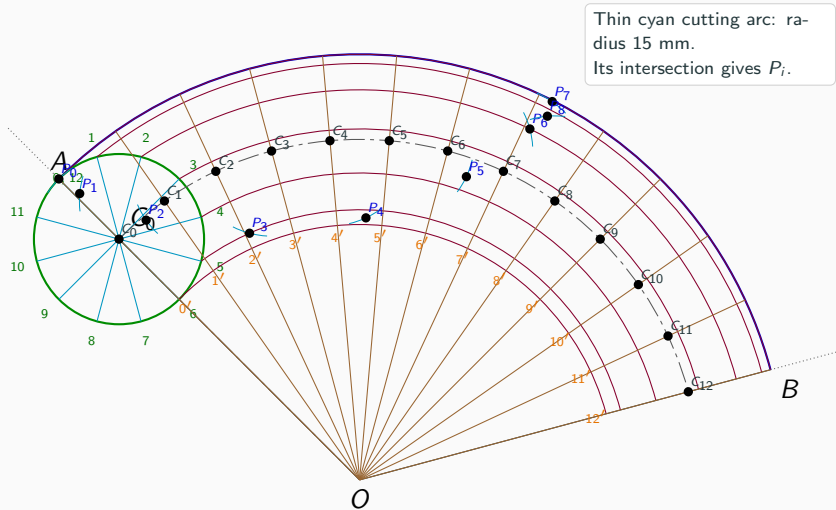
Step 13: Draw the small cutting arc and locate P_i



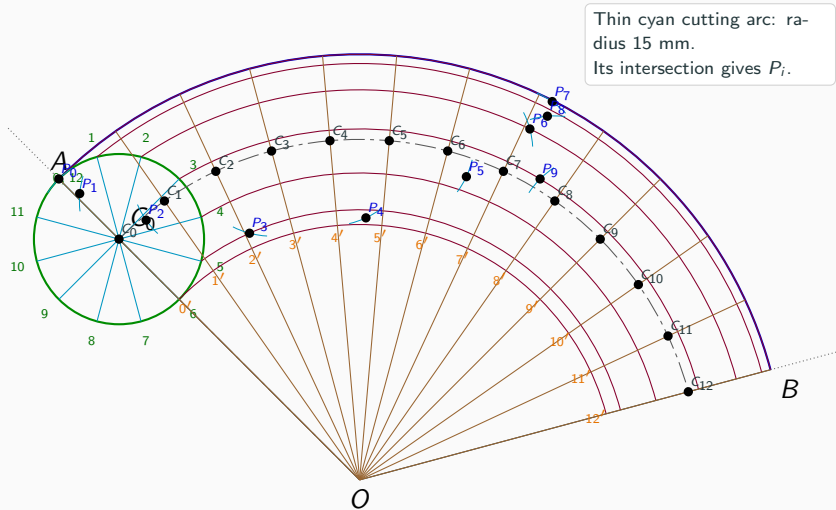
Step 13: Draw the small cutting arc and locate P_i



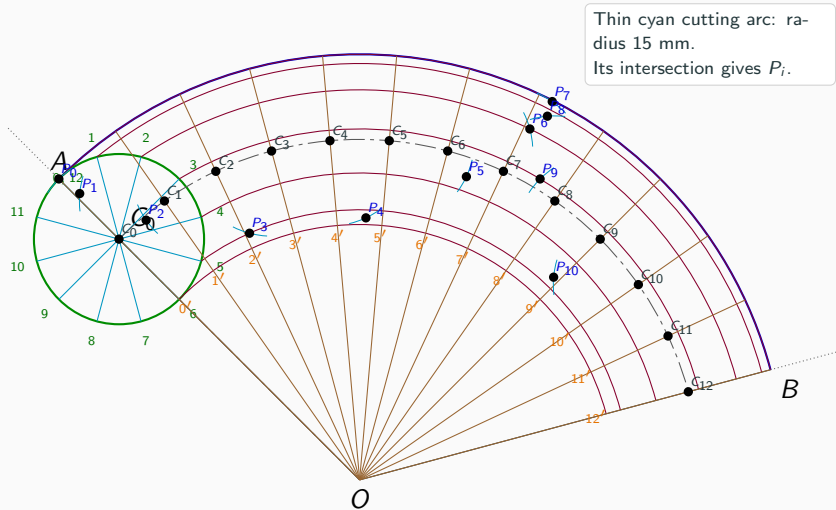
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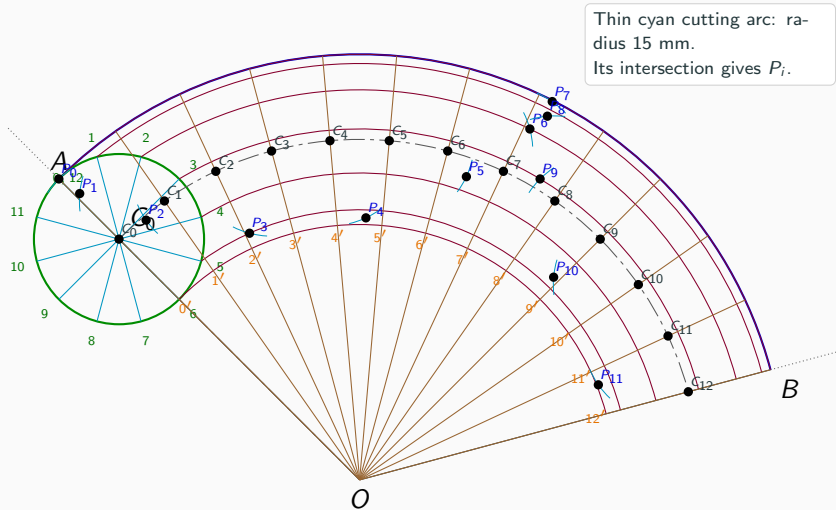
Step 13: Draw the small cutting arc and locate P_i



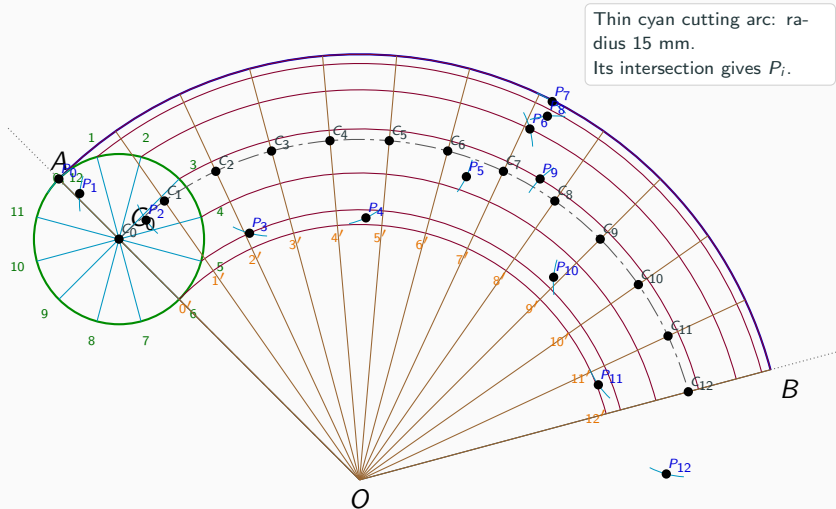
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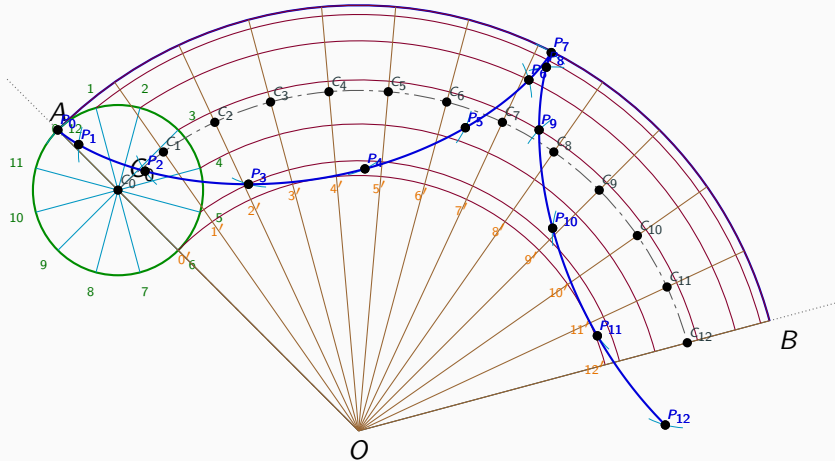


Step 13: Draw the small cutting arc and locate P_i

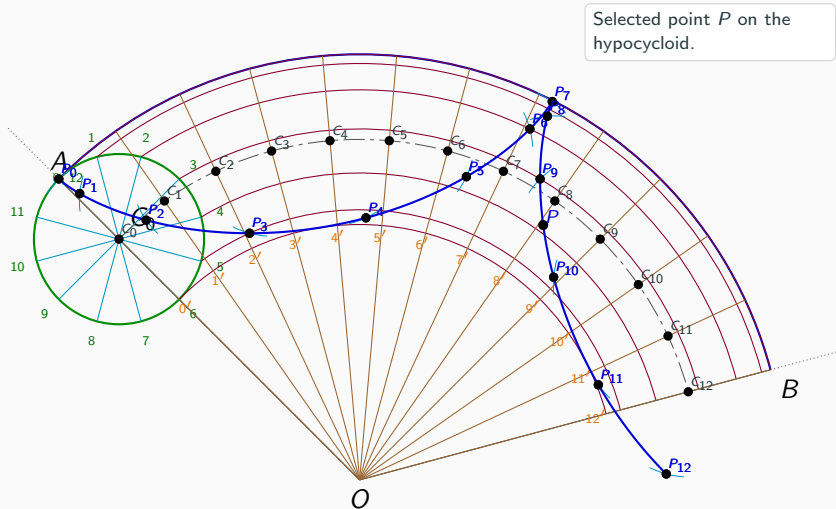


Step 14: Draw the hypocycloid

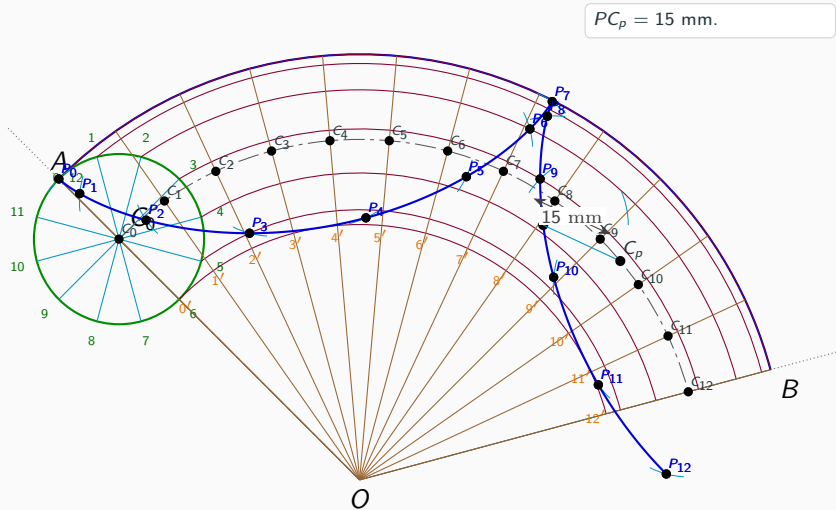
Blue curve: hypocycloid.



Step 15: Select a point P on the hypocycloid

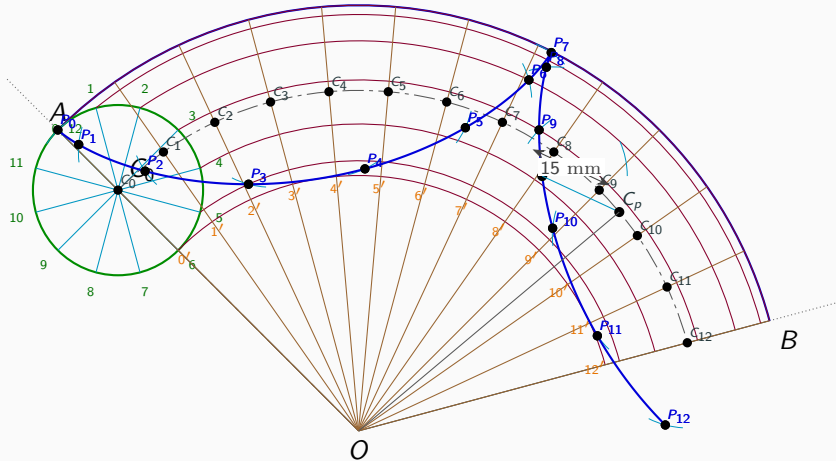


Step 16: Draw the 15 mm arc from P to the centre curve

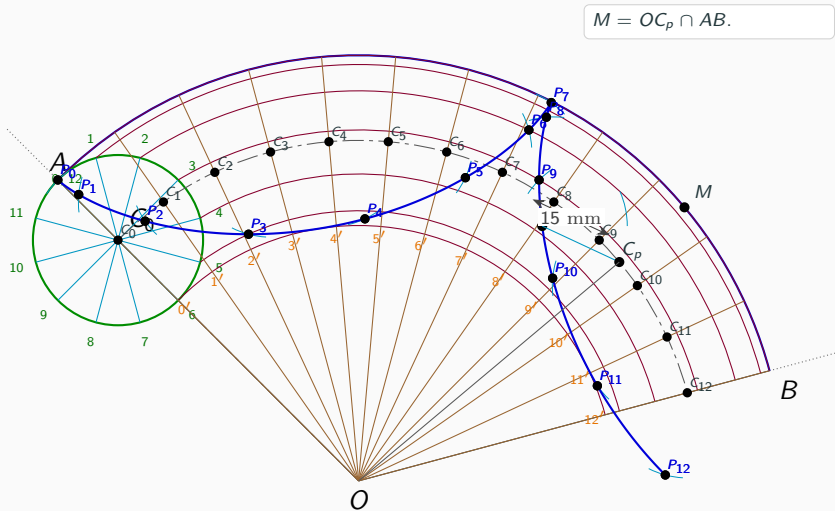


Step 17: Join C_p to O

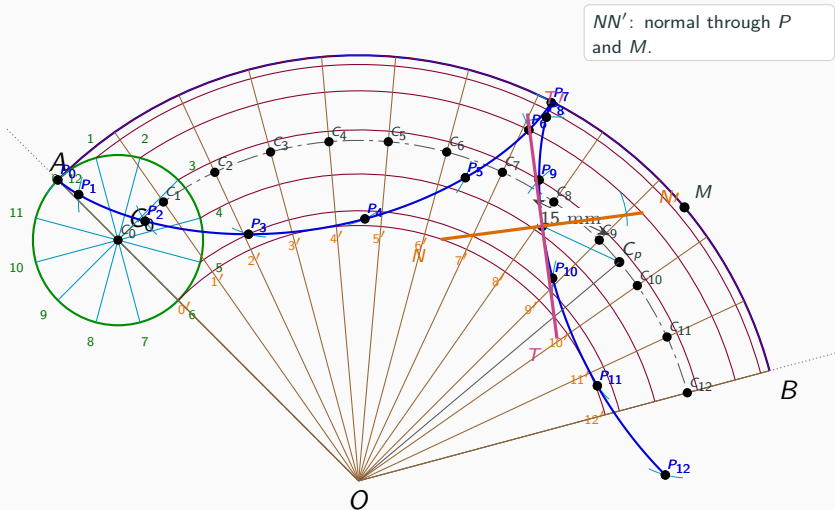
Join C_p to O .



Step 18: Mark M on the directing circle

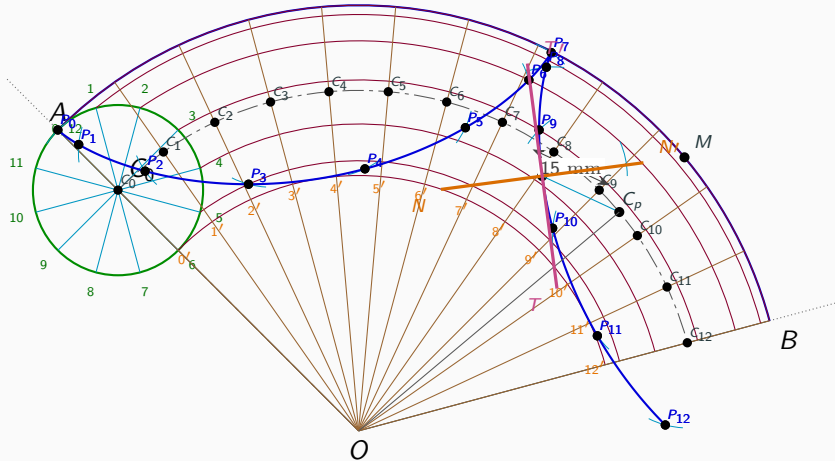


Step 19: Draw the normal NN' through P



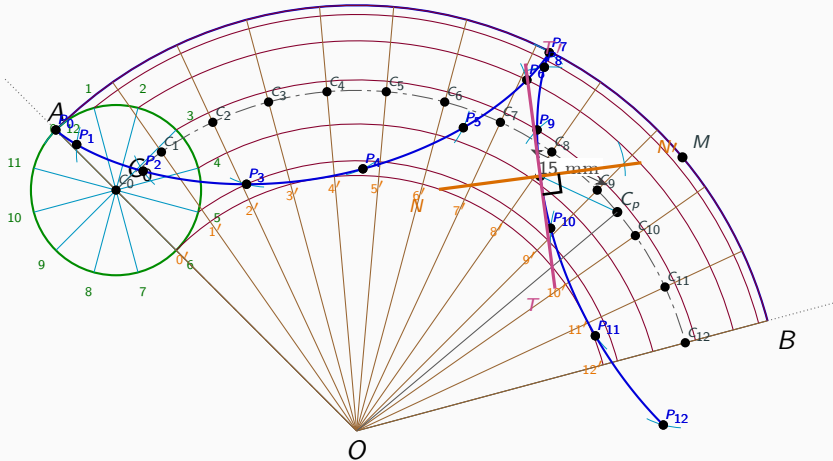
Step 20: Draw the tangent perpendicular to NN'

$$TT' \perp NN'$$



Step 20: Show the right-angle at P

Right angle at P .



Step 20: Final completed hypocycloid construction

